



Uniaxial strain engineering of the electronic and optical properties of phenyl-functionalized black phosphorus

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ABSTRACT

The electronic and optical properties of phenyl-functionalized black phosphorus (phenyl-BP) under uniaxial strain engineering were investigated using first-principles calculations. It revealed that the overall transitions from semiconductor to metallic properties were observed in the phenyl-BP. Notably, when 4% tensile strain was exerted along the armchair direction, phenyl-BP was restored to a p-type semiconductor with a smaller gap of 0.60 eV. Furthermore, compressive strain enhanced its ability to absorb ultraviolet light and high photon energy region around 14 eV. Our results suggested that the strain-induced properties of phenyl-BP provided a theoretical guidance for realizing its full potential in optoelectronic applications.

1. Introduction

Two-dimensional (2D) nanomaterials [1–4], have garnered considerable attention owing to their prominent properties and thus shown widespread applications in energy storage devices, optoelectronic devices and transistors [5,6]. Especially in 2014, the ultra-thin black phosphorus (BP) has been fabricated successfully by the mechanical exfoliation [7], and exhibited immense potential in the photoelectric field [8,9]. BP is regarded as a novel type of p-type 2D semiconductor with a hole mobility of about $1000 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ at room temperature [10]. According to the theoretical research [11], BP has a direct or nearly direct band gap ranging from 0.3 to 2.0 eV, depending on the number of layers or intra-layer strain [12]. Subsequent research identifies that monolayer BP, also known as phosphorene, has a direct band gap value of 1.72 eV [13]. In addition, strain engineering has been widely employed in tuning the electronic properties of 2D materials [14–16]. Compared to other 2D materials, BP exhibits excellent flexibility with an order of magnitude smaller Young's modulus [17]. Peng *et al.* finds that BP can withstand up to 30% of strains without damaging its crystal structure [18]. Particularly, some researches indicate a significant effect of strain engineering on the properties of BP, such as inducing a semiconductor-to-metal transition via strain [19–21].

Unfortunately, at the ambient atmosphere, oxygen and water infiltrated through the interface, then leading to the structural breakdown within the BP layer and seriously affecting its electronic properties [22,23]. Extensive investigations were carried out on covalent

functionalization of BP via aryl as a highly effective and controlled method to eliminate its negative effects [24–26]. In recent years, a growing number of technical fields have begun to explore the effect of covalent functionalization on the BP. For example, phenyl-passivated BP reduced nonradiative electron-hole recombination by a factor of three and effectively limited chemical degradation to three weeks in comparison with the pristine system [27–29]. In addition, functionalized BP was greatly utilized in highly efficient solar cells [30], sodium-ion battery cathode materials [31], and so on. However, a report illustrated that aryl may lead to sublayer P-P bond breakage and dangling bonding of sublayer P atoms [32]. Due to the functionalization, a relatively flat energy band within the band gap was created to result in a jump migration of the local state that caused a reduction in the hole mobility of BP [33]. To address the issues, Li *et al.* reported that surface functionalization of BP via polymers without adversely affecting the electronic transport properties of BP [32]. Moreover, the nitrogen derivative species were predicted by Mou *et al.* to introduce a strong P-N-giving bond at the interface to avoid disturbing its intrinsic electronic structure [34].

Despite these efforts, the potential benefits of strain engineering that strengthened the covalent functionalization of BP have been largely overlooked. Thus, it was of great practical significance to achieve the tunability of a semiconductor with a low band gap. Herein, we in-depth report the effect of uniaxial strain on the structural, electronic, and optical properties of phenyl-functionalized black phosphorus by using first-principles calculations based on the density functional theory.

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Overall, our findings imply the tunability of the photoelectrical properties of phenyl-BP via strain engineering, which provided a theoretical insight for exploring the potential application of functionalized BP in photoelectric devices.

2. Computational details

All first-principles calculations were performed using the density functional theory within the Cambridge Sequential Total Energy Package (CASTEP) code to obtain structural relaxation and properties [35]. Generalized gradient approximation (GGA) with the Perdew, Burke, and Ernzerhof (PBE) function was used to describe the electron exchange–correlation. The empirical DFT-D2 correction proposed by Grimme was employed to eliminate the van der Waals interaction [36]. The sampled Brillouin region in the zone of $3 \times 3 \times 1$ k-points based on a Monkhorst – Pack grid mesh was adopted for geometry optimizations and $5 \times 5 \times 1$ for properties calculations. The cut-off energy was set as 400 eV. Before our properties calculations, all slab models were fully relaxed by employing Broyden-Fletcher-Goldfarb-Shanno (BFGS) method. The energy, force, stress, and displacement tolerances were set as 10^{-5} eV/atom, 0.03 eV/Å, 0.03 GPa, and 0.001 Å, respectively.

To regulate the spacing of surface atoms by adjusting the lattice parameters, the surface electronic structure could be effectively regulated [18]. The calculations applied the tensile and compressive strains to the modeling system along the armchair and zigzag directions. The degree of strain deformation was expressed by ε , which was defined as

$$\varepsilon = (\alpha - \alpha_0) / \alpha_0 \quad (1)$$

Where the α and α_0 were the lattice constants along the single direction for the strained and relaxed structure. Positive values of strain referred to expansion, while negative corresponded to compression. Starting with the phenyl-BP, strains within the range of $\pm 8\%$ were applied in either the ε_x (zigzag) or ε_y (armchair) direction to explore the effects on the geometric, electronic, and optical properties.

3. Results and discussion

As shown in Fig. 1(a), the calculated lattice parameters for intrinsic BP were $a = 3.36$ Å, $b = 4.59$ Å, which was in agreement with experimental values [32,37,38]. The rectangular $4 \times 3 \times 1$ supercell (Fig. 1(c)) was constructed with a vacuum layer of 20 Å set in z direction to eliminate the possible interactions between neighboring slab layers [39]. The total adsorption energy (E_d) was calculated by

$$E_d = E_{BP-org} - E_{BP} - E_{org} \quad (2)$$

Where E_{BP-org} , E_{BP} , and E_{org} represented the total energy of functionalized BP, the monolayer BP and the isolated organic groups. A positive E_d indicated a thermodynamic instability of covalent functionalization, while a negative E_d suggested thermodynamic stability. In this work, the phenyl was selected as the covalently functionalized group, with the benzene ring having three possible highly symmetric adsorption orientations to the BP surface, i.e., armchair, zigzag, or slant orientations as seen from Fig. 1(b). The adsorption energy for each position was displayed in Table 1, in which the armchair was the most stable of all the position considered. The lone pair electrons of surface P atoms redirected to the downside and newly formed P-C bond replaced previously existed P-P bond (Fig. 1(c, d)). To further analyze the stability of the functionalized structure, Fig. 2(a) depicted the phonon dispersion bands of phenyl-BP to evaluate dynamical stability. Obviously, the phonon branches did not contain imaginary frequencies that confirmed the dynamical stability of phenyl-BP. As exhibited in Fig. 2(b), we then examined changes in the potential energy from the molecular dynamics simulation at 300 K in 10 ps. The potential energy fluctuated within a smaller range, suggesting that phenyl-BP existed stably and there would be no deformation under ambient conditions.

Table 1
Adsorption energies (E_d) of phenyl groups along different orientations.

Type	armchair	zigzag	slant
E_d (eV)	−2.06	−1.96	−1.99

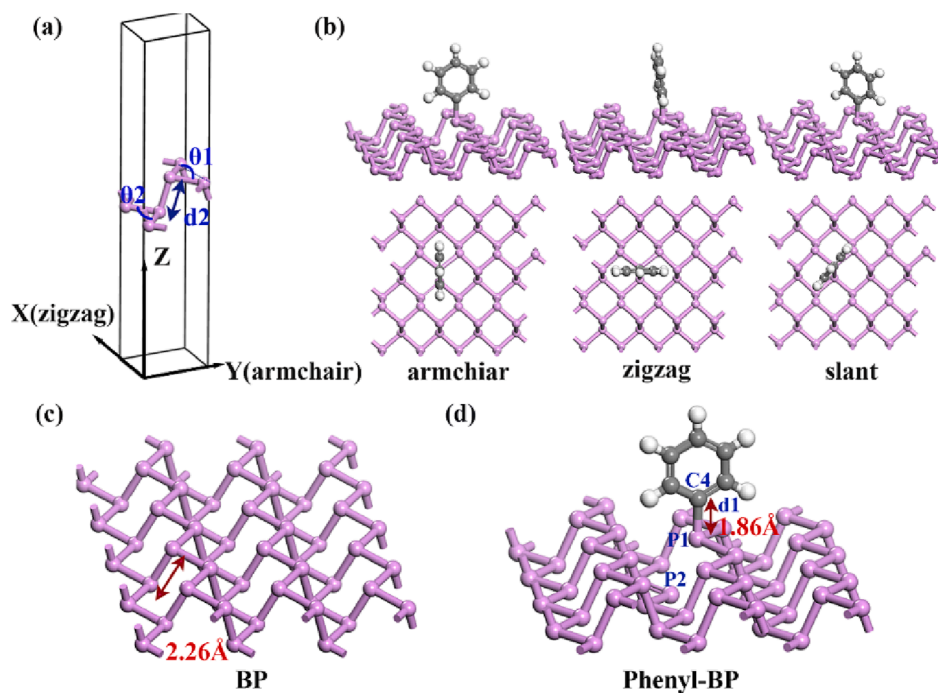


Fig. 1. (a) Atomic structure of monolayer BP. (b) Three possible orientations of the phenyl on the BP surface. (c) The structure of the intrinsic BP. (d) Geometry and bond distances of the most stable functionalized BP along the armchair direction. The white, gray and purple balls represent H, C and P atoms, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

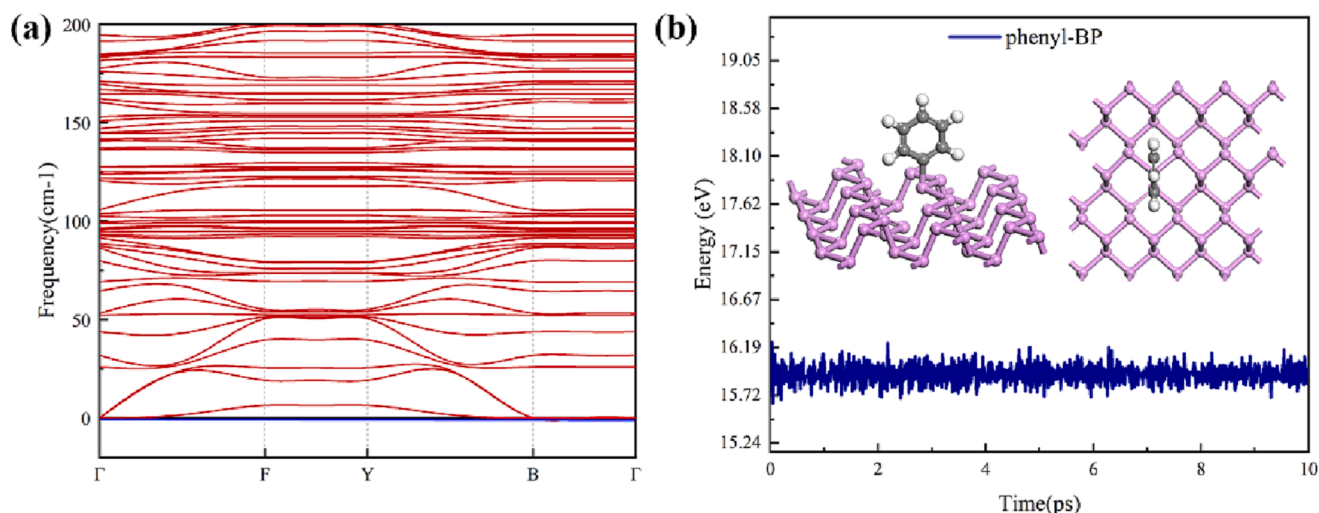


Fig. 2. (a) Phonon dispersions and (b) the changes in potential energy in molecular dynamics simulation (300 K, 10 ps) for phenyl-BP.

The calculated geometrical parameters and adsorption energy (E_d) under uniaxial strain were presented in Table 2 (d1, d2, θ_1 , and θ_2 are labeled in Fig. 1(a, d)). The E_d , P-P and P-C bond lengths of phenyl were -2.05 eV, 2.80 Å and 1.86 Å, which was nearly consistent with that of Mou *et al.* [34]. Both tensile and compressive strain subtly affected the d1 and d2 bond lengths. The adjacent angle θ_1 and the interlayer angle θ_2 increased with increasing tensile strain. Along the armchair direction, the E_d increased from -2.06 eV to -0.19 eV under tensile strain, while the effect of compressive strain showed irregular increase. The E_d increased monotonically in two directions after applying strain along the zigzag direction, then the value turned positive and started to become unstable with the rise of strain. This indicated that strain applied along the armchair direction were more practical than the zigzag direction. Significantly, d2 changed abruptly at 8% tensile strain along the zigzag direction and restored to an approximate length about intrinsic BP (2.23 Å). These structural changes caused by strain greatly affected the electronic properties of phenyl-BP.

The band structure and partial density of state (PDOS) of monolayer BP were displayed in Fig. 3(a, c). The results testified that BP was a direct band gap semiconductor with a value of 0.99 eV at Γ -point, which was consistent with previous reports [18,40]. Fig. 3(b) depicted the mid-gap state indicated by the red line due to the generation of dangling bonds between sub-layer P atoms. It was clear that the mid-gap of phenyl-BP overlapped with the valence band and crossed the Fermi energy, revealing the transitions from semiconductor to metallic properties. The conduction and valence bands exhibited a general downward shift due to the decrease in energy level. According to the PDOS analysis, it was found that intrinsic BP had a wide gap near the Fermi energy, which corresponded to the band gap and had a strong *sp* orbital hybridization. Obviously, the p orbital had a relatively larger contribution to both conduction and valence bands. The phenyl-BP had split waves

across the Fermi energy level, which were mainly from the orbital hybridization of the p orbital in the P atom with the C atom of the phenyl. Additionally, the Fermi energy level started to be filled with electrons, leading to the loss of band gap.

Fig. 4 displayed the band structures and PDOS of P atoms under four different strains along two directions: The first set of strains (4%, 8%, -4% and -8%) were applied along the armchair direction, while the second set (4%, 8%, -4% and -8%) were applied along the zigzag direction. Noteworthy, as shown in Fig. 4(a), the band gap of phenyl-BP was opened and recovered to a p-type indirect band gap semiconductor when 4% tensile strain was applied. The highest point of the valence band shifted to the F-point, and the band gap was reduced to 0.60 eV. Under the influence of strain, the electrons that were originally located in the mid-gap state transferred to the nearest conduction band and left holes, thereby altering the conduction characteristics of phenyl-BP. While other strains induced phenyl-BP to maintained metallic properties and the electrons were more likely to cross the Fermi energy level with the rise of strain. Regarding the PDOS, there was still more contribution from the p-orbital of the P atom in the mid-gap state. Fig. 4(e-h) depicted the effect of exerting strains on the phenyl-BP along the zigzag direction. The overall change was similar to the direction along the armchair. When the strain reached 8%, it was found that the mid-gap state became distorted and the conduction band crossed the Fermi energy level. To further understand the interlayer interactions under strain, Fig. 5 manifested the electron density difference of strain-free phenyl-BP and 8% strain system along the zigzag direction. The electron density distribution on phenyl remained similar (Fig. 5(a, b)). Surprisingly, we observed significant differences in the interlayer P atom associated with functionalization (Fig. 5(c, d)). Strain caused a lone electronic interaction around the broken P-P bond, leading to the disappearance of suspended bonds of the sublayer P atom. This resulted in not only remained covalently functionalization with phenyl but also had a bonding state with the interlayer P atom under strain like intrinsic BP. The analysis was consistent with the geometrical parameters which was depicted in Table 1.

It was well known that the degree of electron sharing between two atoms were evaluated using the overlap population, which reflected the covalent or ionic nature of a chemical bond. A higher value of overlap population represented a higher covalence level. The Mulliken's population analysis of the charge transfer and overlap population between black phosphorus and phenyl under uniaxial strain were illustrated in Table 3 [40–42]. As displayed in Fig. 1(d), P1, P2, and C4 represented the atoms that underwent considerable changes in the modified region. The overlap population of P1-P2 in BP was about 0.45 , indicating a

Table 2

Calculated structural parameters (P-C and P-P bond lengths d1, d2; bond, angles θ_1 , θ_2) and adsorption E_d of phenyl-BP under strain.

Direction	Strain (%)	E_d (eV)	d1 (Å)	d2 (Å)	θ_1 (°)	θ_2 (°)
ϵ_x	-8	0.22	1.86	2.91	95.27	98.01
	-4	-2.02	1.86	2.83	98.53	99.67
	0	-2.06	1.86	2.80	100.55	101.68
	4	-0.39	1.86	2.77	104.82	101.57
	8	2.12	1.86	2.23	118.56	114.69
ϵ_y	-8	-1.64	1.86	2.99	98.91	89.35
	-4	-2.08	1.86	2.83	100.99	96.90
	4	-1.54	1.86	2.84	104.18	103.27
	8	-0.70	1.86	2.81	105.65	110.30

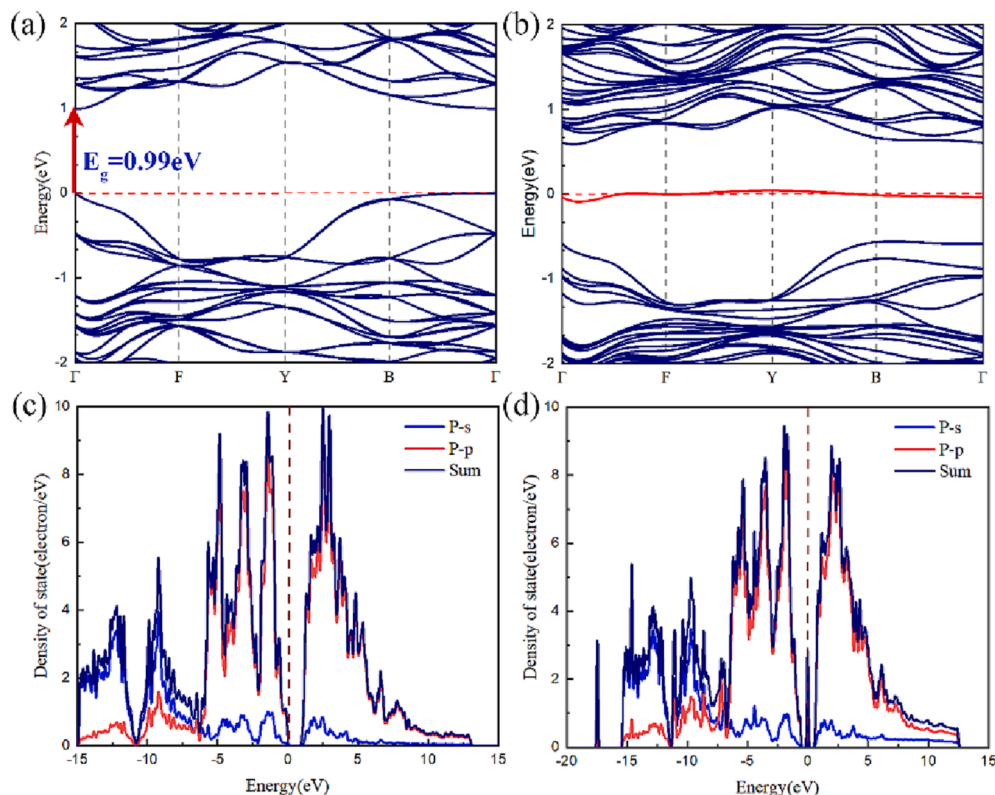


Fig. 3. (a) The Band structure and (c) PDOS of monolayer BP. (b) The band structure and (d) PDOS of phenyl-BP.

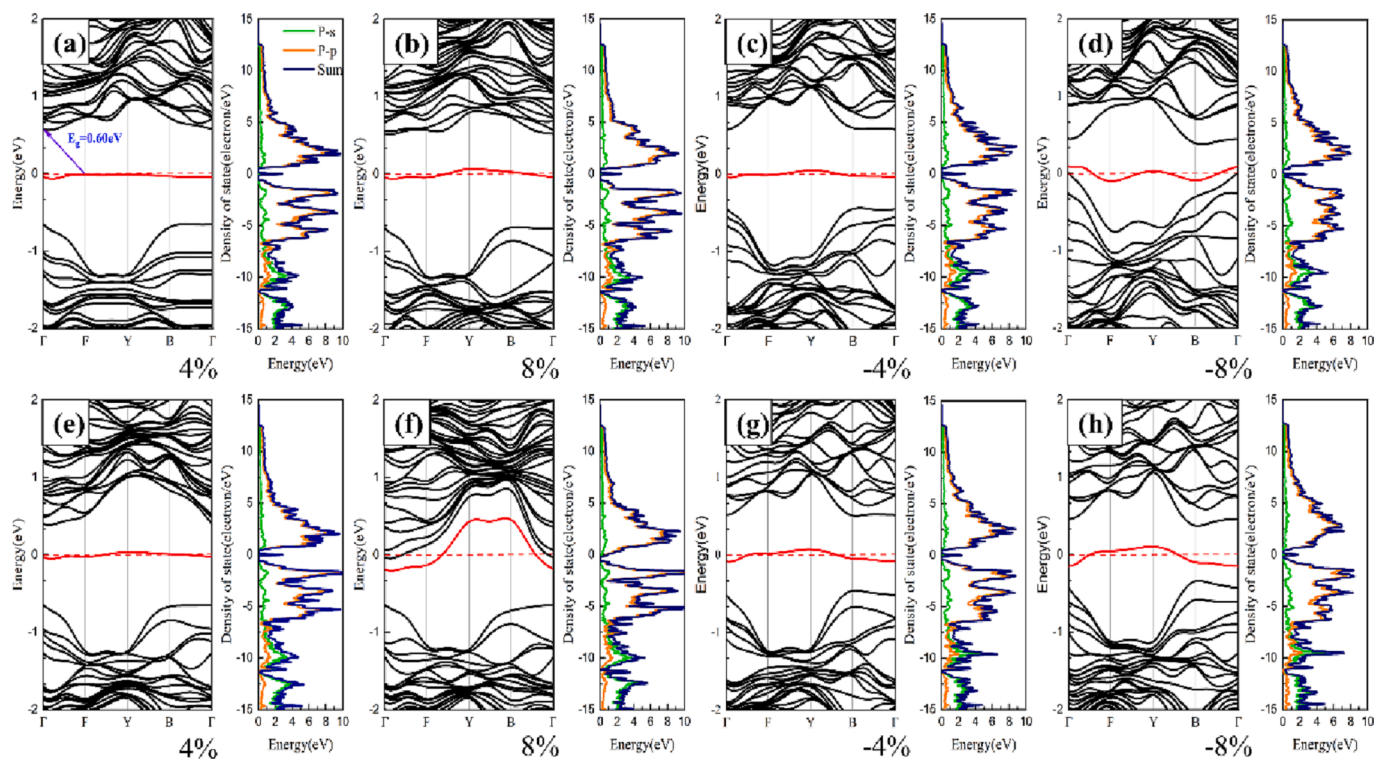


Fig. 4. Band structures and the partial density of states of phenyl-BP under different strains. (a) to (d) represented the strains of 4%, 8%, -4% and -8% along the armchair direction. (e) to (h) represented the strains of 4%, 8%, -4% and -8% along the zigzag direction.

connection through covalent bonds. However, the value decreased to 0.01 after covalent functionalization, existing a very weak antibonding (or non-bonding state). The population of C4-P1 was 0.53, indicating a

stronger covalence than that of BP. Interestingly, most of the strains did not alter this bonding feature. However, when 8% strain was applied along the zigzag direction, the overlap population of previously broken

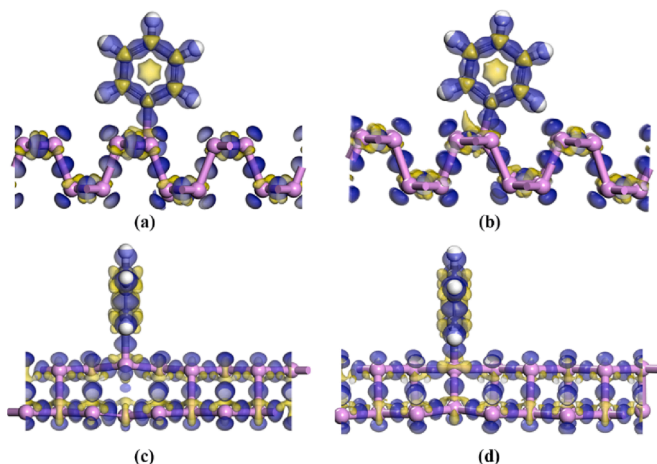


Fig. 5. The electronic density difference of (a, c) phenyl-BP without strain and (b, d) 8% strain system. The isosurface value was taken as $0.07 \text{ e}/\text{\AA}^3$.

Table 3

Mulliken's population analysis and charge transfer between BP and phenyl in different directions and strains.

Direction	Strain (%)	Bond	Overlap population	Charge transfer (e)
ϵ_x	-8	C4 - P1	0.59	0.15
		P1 - P2	0.01	
	-4	C4 - P1	0.57	0.17
		P1 - P2	-0.01	
	0	C4 - P1	0.55	0.15
		P1 - P2	-0.03	
	4	C4 - P1	0.55	0.12
		P1 - P2	-0.03	
	8	C4 - P1	0.63	0.09
		P1 - P2	0.45	
ϵ_y	-8	C4 - P1	0.57	0.18
		P1 - P2	-0.01	
	-4	C4 - P1	0.56	0.15
		P1 - P2	-0.01	
	4	C4 - P1	0.56	0.11
		P1 - P2	-0.05	
	8	C4 - P1	0.60	0.10
		P1 - P2	-0.07	

P1-P2 bond recovered to 0.45, which in accordance with the previous analysis of the electron density difference. Besides, the population of the C4-P1 reached 0.63, thereby enhancing the covalent bond strength. During the functionalization, including a dipole correction in the calculation, a total of 0.15 e of charge transfer was generated, which was expressed as the phenyl presenting electrons from the BP. An increase in

tensile strain reduced the charge transfer by 0.05–0.06 e whereas the compressive strain remained the charge transfer roughly constant.

Numerous works have discussed the optical properties of black phosphorus [20,43,44]. Motivated by some analyses of previous theories, the optical properties of phenyl-BP under uniaxial strain were investigated. Fig. 6 showed the optical reflectivity of intrinsic BP and phenyl-BP under different strains along the x-axis and y-axis of polarized light. Fig. 6(a, b) depicted the reflectivity of phenyl-BP under strain along the armchair direction. There were three main reflection peaks along the x-axis without strain, which occurred at approximately 0, 4.36, and 8.34 eV. A maximum reflectivity occurred at an incident photon energy of 8.34 eV along the y-axis. BP under strain along the zigzag direction (Fig. 6(c, d)) had two reflection peaks near 1.33 and 8.27 eV along the x-axis. When the energy of incident light reached 8.27 eV, the reflectivity got the maximum of 0.30. In contrast to the phenomenon observed along the x-axis, the overall reflectivity was lower along the y-axis. Besides, the phenyl-BP under tensile strain produced two approximate reflectivity peaks near the original peak. When the incident photon energy exceeded 25 eV, the reflectivity tended to zero in two directions under all strains. As the strain increased, compressive and tensile strains showed opposite patterns of variation in which the peak height increased with compression strain, moving towards higher energy, while the opposite was observed for tensile strain. The phenyl-BP mainly reflected in the photon energy region of 5–10 eV after functionalization. In the visible light region ($<3 \text{ eV}$), compressive strain along the armchair direction increased the reflection efficiency of BP.

The absorption spectra of BP and phenyl-BP under different strains were plotted in Fig. 7. A broad absorption edge of intrinsic BP over the 2–20 eV region was clearly observed without strain (the black line in Fig. 7(a)). When the photon energy exceeded 20 eV, BP stopped absorbing the injected light. The energy of the absorption edge (around 14 eV) was improved by functionalization and strain engineering, resulting in a blueshift phenomenon and enhancing the absorption of ultraviolet light in phenyl-BP. As shown in Fig. 7(a, b), the adsorption edges of pure, 0%, -8%, -4%, 4%, and 8% strained systems became 1.0, 0.52, 0.44, 0.23, 0.46, and 0.56 eV along x-axis. A redshift phenomenon was obtained in the visible light region, which reached its maximum when 8% compressive strain was applied. When the polarized lights were introduced along the zigzag strain direction (Fig. 7(c, d)), the absorption peaks decreased and shifted to the lower energies. The polarized light along the y-axis showed increasing adsorption edges around 2.84 to 3.83 eV. The adsorption edges were at 0.96, 0.54, 0.23, 0.18, 0.48, and 0.54 eV for the pure, 0%, -8%, -4%, 4%, and 8% strained systems along x-axis. With an increase of strain, the intensity of the optical absorption was heightened and the absorption region became more expansive simultaneously, proving that the system was easily excited under the influence of compressive strain. More importantly, both reflectivity and adsorption manifested a sensitivity to strain, among

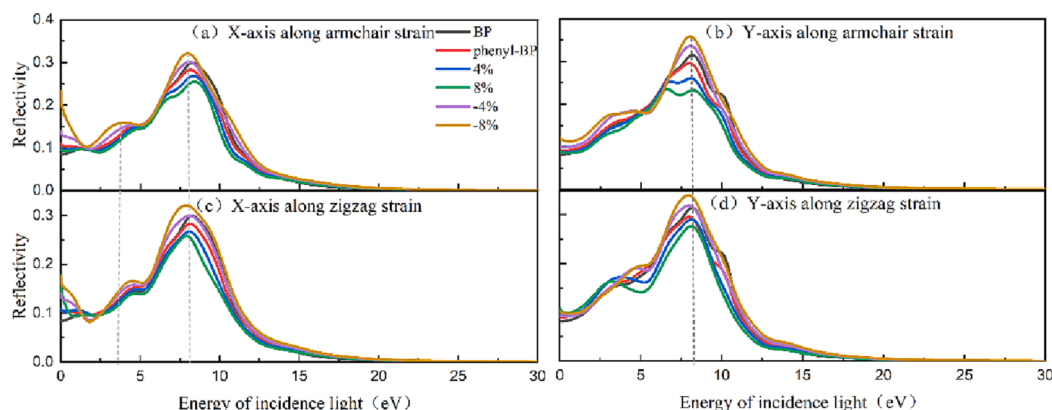


Fig. 6. The reflectivity of BP and phenyl-BP as a function of strain.

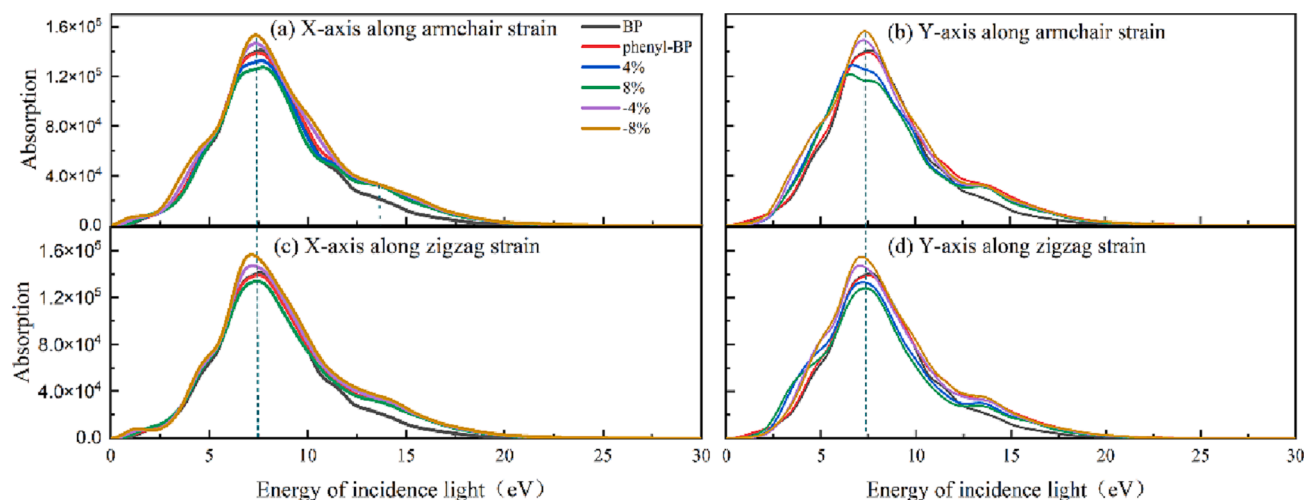


Fig. 7. Absorption spectra of BP and phenyl-BP as a function of strain.

them, compressive strain caused the peaks to rise while the tensile strain was the opposite. Moreover, the reflectivity and adsorption of phenyl-BP revealed the fact that both x- and y-axes exhibited a strong anisotropy. Therefore, an effective way was provided to regulate the optical properties of phenyl-BP, especially in applications such as the ultraviolet absorption materials.

4. Conclusion

To summarize, by performing first-principles calculations, we detailedly studied the geometric, electronic and optical properties of phenyl-functionalized BP by means of uniaxial strain engineering. After covalent functionalization, the valence band of phenyl-BP crossed the Fermi energy level, thereby resulting in obvious transitions from semiconductor to metallic properties. When 4% tensile strain was applied along the armchair direction, phenyl-BP was adjusted to a p-type semiconductor with a smaller indirect band gap. Interestingly, the tensile strain had a more profound effect on the electronic properties of phenyl-BP. Besides, a significant optical anisotropy of phenyl-BP was manifested under different strains and polarized light directions. The absorption and reflection peaks of phenyl-BP decreased, which caused that the intensity of optical properties was reduced to a certain extent. In the visible light regions, compressive strain along the armchair direction increased the reflection efficiency of phenyl-BP. The intensity of the optical absorption region around 14 eV was heightened, which was particularly for absorbing ultraviolet light and high photon energy. We demonstrate that strain engineering achieves the tunability of the photoelectrical properties of phenyl-functionalized black phosphorus; thus, it offered a theoretical guidance for making it a candidate for future nanoelectronic and optoelectronic applications.

CRediT authorship contribution statement

Zilin Chen: Methodology, Formal analysis, Data curation, Writing – original draft, Writing – review & editing. **Huadong Zeng:** Conceptualization, Validation, Formal analysis, Resources, Investigation, Writing – review & editing, Visualization, Supervision, Funding acquisition. **Houyu Chen:** Data curation, Writing – review & editing. **Longying Wen:** Methodology, Validation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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